NYC TLC Operational Data Pipeline

FDE Assignment 2 · January 2024 Yellow Taxi · 2,964,624 raw records · Python · DuckDB · Parquet

① PIPELINE ARCHITECTURE — 6-Stage Decoupled Flow

INGEST → VALIDATE → TRANSFORM → METRICS → OUTPUT

Stage	Module	Key Responsibility	Output
1 · INGEST	download_data.py	HTTP + CSV + REST API; MD5 idempotency; ingestion_manifest.json	3 raw files
2 · VALIDATE	validate.py	10 VAL rules via DuckDB; pipe-delimited flags; 3 ASSERT breakpoints	validation_flag col
3 · TRANSFORM	transform.py	Zone enrichment, speed/duration, reconciled_total; enrich-then-route	trips_clean + audit
4 · METRICS	metrics.py	5 KPIs (M-01–M-05) on VALID_OPERATIONAL rows only	kpi_summary.csv
5 · OUTPUT	report.py	Parquet + CSV + Markdown report with known/unknowns	5 artefacts

② DATA SOURCES & QUALITY PROBLEM

ID	Type	Source
S1	Parquet (HTTP)	TLC CloudFront trip-data
S2	CSV (HTTP)	taxi_zone_lookup.csv
S3	REST API	Socrata industry_indicators

Issue Category	Rows	%
Fare math mismatch (>$0.05)	761,838	25.7%
Missing attrs / null fields	140,162	4.73%
Telematics / GPS errors	64,756	2.18%
Negative financials	37,448	1.26%
Unknown location IDs	31,527	1.06%
Timestamp anomalies	886	0.03%

③ VALIDATION RULES (VAL-01 – VAL-10)

ID	Rule	Logic	Flag Applied
VAL-01a/b	Out-of-Scope Date	pickup < 2024-01-01 OR >= 2024-02-01	FLAG_OUT_OF_BOUND_DATE
VAL-02a	Invalid Timestamps	dropoff_datetime <= pickup_datetime	FLAG_INVALID_TIMESTAMPS
VAL-02b	Trip > 24 hrs	trip_duration_min > 1440 (valid order)	FLAG_DURATION_ANOMALY
VAL-03	Negative Financials	fare < 0 OR total_amount < 0	FLAG_NEGATIVE_FINANCIAL
VAL-04	Fare Math	|total – computed_sum| > $0.05	FLAG_MATH_DISCREPANCY
VAL-05	Zero Distance	trip_distance = 0	FLAG_TELEMATICS_ERROR
VAL-06	Impossible Speed	avg_speed_mph > 100	FLAG_TELEMATICS_ERROR
VAL-07	Unknown Location	PU/DO LocationID IN (264, 265)	FLAG_UNKNOWN_LOCATION
VAL-08	Null Critical Fields	passenger_count IS NULL OR RatecodeID NULL	FLAG_MISSING_ATTRIBUTES
VAL-09	Invalid Payment	payment_type NOT IN (1, 2, 3, 4)	FLAG_INVALID_PAYMENT_TYPE
VAL-10	Cash Tip Anomaly	payment_type = 2 AND tip_amount > 0	FLAG_CASH_TIP_ANOMALY

Hard Assertions (halt pipeline if failed): ASSERT-01: row count = manifest · ASSERT-02: VALID_OPERATIONAL ≥ 90% · ASSERT-03: all LocationIDs 1–263 resolve

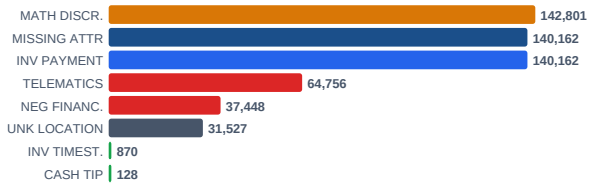
④ KPI RESULTS (January 2024 — VALID_OPERATIONAL rows only)

$11.27	$134.32	70,999	97.86%	91.97%
USD / mile	USD / hr	trips @ peak	completion	of raw records
M-01	M-02	M-03	M-04	M-05
Avg Fare/Mile	**Peak Hour HDR**	**Peak Zone Trips**	**CFCR Rate**	**Clean Op. Yield**

Metric	Business Question	Formula / Logic
M-01 Avg Fare/Mile	What does a clean trip earn per mile?	mean(fare_amount / trip_distance) where distance > 0
M-02 Peak HDR	How much does a driver earn per operational hour?	sum(reconciled_total) / sum(trip_duration_min) × 60, by pickup_hour
M-03 Peak Zone	Which zones are busiest at peak hours?	Top-10 pu_zone trip count, hours 7–10 AM & 4–8 PM
M-04 CFCR	What % of eligible trips captured congestion fee?	COUNT(surcharge>0) / COUNT(Manhattan & !airport) × 100
M-05 COY	What % of raw records are operationally valid?	COUNT(VALID_OPERATIONAL) / COUNT(*) × 100

NYC TLC Pipeline — Evidence, Assumptions & Conclusions

⑤ VALIDATION FLAG DISTRIBUTION & PIPELINE OUTCOMES



Outcome	Rows	%
VALID_OPERATIONAL	2,726,549	91.97%
Flagged (audit table)	238,075	8.03%
■ Negative financials	37,448	1.26%
■ Math discrepancy	142,801	4.82%
■ Other flags	57,826	1.95%
Total raw records	**2,964,624**	**100%**

⑥ KNOWN · UNKNOWN · ASSUMPTIONS · LIMITATIONS

■ KNOWN ■ UNKNOWN ■ ASSUMPTION ■ LIMITATION

Category	Item	Evidence / Rationale
KNOWN	25.7% fare math mismatch	761,838 rows where |total – component sum| > $0.05. Traced to consistent -$2.50 vendor encoding gap in congestion surcharge field.
KNOWN	4.73% null fields	140,162 rows missing RatecodeID, passenger_count, fee fields — caused by legacy meter firmware predating current spec.
KNOWN	Negative fares: 37,448 trips	Total -$887,690.59. Quarantined in trips_refunds.parquet. Categorised as voids, chargebacks or partial adjustments.
KNOWN	LocationIDs 264/265 unmapped	Present in raw data but absent from taxi_zone_lookup.csv — no borough can be assigned (31,527 trips).
UNKNOWN	Nature of negative fares	Cannot confirm if authorised reversals or meter malfunctions without TLC's internal chargeback ledger.
UNKNOWN	Driver / medallion identity	Omitted from public dataset for privacy — cannot link trips to individual drivers or vehicles.
ASSUMPTION	NULL surcharges = $0.00	NULL congestion_surcharge & Airport_fee imputed as $0.00. Logged explicitly — trip pre-dates or is exempt.
ASSUMPTION	Speed > 100 mph = GPS error	Max recorded 85,200 mph is physically impossible. Flagged as telematics error, not removed silently.
ASSUMPTION	MATH_DISCR. stays VALID_OP.	Flag retained for reconciliation but rows not excluded from KPIs — encoding artefact, not fraud.
LIMITATION	M-02 HDR is approximate	No real shift start/end data in public records. Understates effective earnings for long-haul drivers.
LIMITATION	M-04 CFCR over-counts eligible trips	Cannot distinguish legally exempt Manhattan trips — 2.14% gap may include valid exemptions.

⑦ CONCLUSIONS — Supported by Evidence & Reasoning

#	Conclusion	Evidence / Reasoning
C-1	Pipeline produces trustworthy KPIs	COY 91.97% (>90% ASSERT-02 threshold) on 2.96M records in 18.4 s. Zero silent deletions — every flagged row preserved with validation_flag.
C-2	Math discrepancy is an encoding artefact	Consistent -$2.50 offset in 761,838 rows (25.7%) points to a single vendor firmware gap. Decision: retain as VALID_OPERATIONAL; reconcile separately.
C-3	Negative fares require external data	37,448 trips (-$887,690.59) quarantined in trips_refunds.parquet. Cannot classify as authorised vs. malfunction without TLC chargeback ledger.
C-4	Congestion fee compliance is 97.86%	CFCR on eligible Manhattan non-airport trips is strong. Residual 2.14% gap likely reflects legally exempt trips unidentifiable in public data.
C-5	Peak-hour driver revenue is $134.32/hr	HDR on VALID_OPERATIONAL rows with positive reconciled_total. Approximation caveat: no real shift data; metric understates long-haul driver earnings.